

Reviewing Data Collection Methods

Data Collection and Preprocessing

1. Data Collection

- Data collection refers to the systematic process of **gathering information** from different sources to answer research questions, support decision-making, or measure performance.

Common data collection methods include:

Primary Data Collection

Data collected directly from the field.

- **Surveys & Questionnaires** (online/physical)
- **Interviews** (structured/unstructured)
- **Observations**
- **Experiments**
- **Mobile data collection tools** (e.g., ODK, KoboToolbox, mWater)

Data Collection and Preprocessing

Secondary Data Collection

Using existing data that was collected for another purpose.

- Government databases
- Organizational records
- Academic journals
- Administrative data systems
- Big data sources (social media, sensors, logs)

2. Preprocessing During Data Collection

Preprocessing at the point of collection ensures that the data enters the system **clean, standardized, and usable**. This reduces the cleaning workload later.

Key preprocessing activities during collection include:

- **Designing standardized forms**

Ensures uniform data entry across respondents.

- **Setting validation rules**

E.g. age must be between 0 and 120, phone numbers must be 10 digits.

- **Using mandatory fields**

Essential data cannot be skipped.

2. Preprocessing During Data Collection

- **Skip logic / branching**

Respondents only see relevant questions.

- **Automatic data coding**

E.g., Male = 1, Female = 2; Yes = 1, No = 0.

- **GPS and timestamp auto-capture**

Ensures location and time accuracy.

- **Avoiding free text fields**

Using dropdowns and predefined categories to reduce entry errors.

- **Real-time data quality checks**

Mobile tools can flag missing values or unusual entries immediately.

Preprocessing at collection ensures **accuracy, completeness, and consistency** before the data even reaches the cleaning phase.

b. Data Cleaning and Preprocessing

Once the data is collected, it typically has issues.

Data cleaning and **preprocessing** improve data quality before analysis or modeling.

1. Data Cleaning

This focuses on correcting errors and making the dataset consistent.

Main tasks include:

Handling Missing Data

- Delete rows with excessive missing values
- Replace (impute) missing values with mean/median/mode
- Use interpolation (for time-series data)

b. Data Cleaning and Preprocessing

Handling Outliers

- Identify extreme values using boxplots or z-scores
- Remove or cap them depending on context

Correcting Errors

- Fix incorrect spellings
- Correct inconsistent units (e.g., meters vs. centimeters)
- Remove duplicate entries

Data Type Corrections

- Ensure numeric fields are numeric, not text
- Convert dates to proper date format

Dealing With Inconsistent Entries

Examples:

- “Male,” “M,” and “m” should be standardized
- “Lilongwe,” “LLW,” “Llw” should be harmonized

2. Data Preprocessing (after cleaning)

This prepares data for analysis or modeling.

Key activities:

Data Transformation

- Normalization (scaling values to a fixed range)
- Standardization (z-score transformation)
- Log transformation (to reduce skewness)

Feature Engineering

- Creating new variables from existing ones
e.g., age groups, ratios, percentages.

2. Data Preprocessing (after cleaning)

Encoding Categorical Data

- One-hot encoding (dummy variables)
- Label encoding

Discretization (Binning)

- Converting continuous data into categories
e.g., Age 0–17 = Child; 18–35 = Youth, etc.

Integration

- Combining multiple datasets
(e.g., merging survey data with administrative records)

Data Reduction

- Removing irrelevant variables
- Principal Component Analysis (PCA) for dimensionality reduction